IBVAP — Project Strengths & Honest Technical Limitations

14. SIH Competitive Advantages & Strategic Future Scope

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1. Key Project Strengths (Ranked for SIH Judges)

1. End-to-End Complete Ecosystem (`STRENGTH #1`)

IBVAP is not just a Jupyter Notebook script or a standalone AI model. It provides a complete, working pipeline spanning video ingestion, neural detection, multi-object tracking, spatial zone logic, evidence extraction, cloud database persistence, SHA-256 cryptographic logging, and an executive web command center.

2. Cryptographic Tamper-Evident Audit Logging (`STRENGTH #2`)

While competing hackathon solutions rely on plain SQL or CSV logs that can easily be edited or deleted, IBVAP incorporates SHA-256 linear hash chaining (`secure_audit_log.csv`) and block-level verification (`blockchain_ledger.json`), offering verifiable forensic security.

3. Bulletproof Multi-Tier Resilience (`STRENGTH #3`)

The automatic 3-tier data fallback (MongoDB Cloud $\rightarrow$ Local CSV $\rightarrow$ Seed Data) guarantees that the platform will **never crash** during live judge evaluations, even in high-latency or offline environments.

4. Cutting-Edge Computer Vision Integration (`STRENGTH #4`)

Utilizes Ultralytics YOLO11 paired with ByteTrack (high/low confidence score fusion and Kalman filtering) to minimize track loss and duplicate alerts.

5. Premium Modern UI/UX (`STRENGTH #5`)

Custom dark navy theme, glassmorphic styling, responsive layout, dynamic status indicators, and interactive Plotly visualization charts tailored for defense command centers.

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2. Honest Technical Limitations & Realistic Mitigations

Technical Limitation	Honest Reality in Current Code	Proposed Future Improvement
Single Video Stream Ingestion	Currently processes one video stream (`video.mp4` / `CAM-01`) at a time.	Implement multi-threaded RTSP video stream ingestion with a Redis message queue.

Model Precision / FPS Unmeasured	Quantitative benchmark metrics (mAP, FPS, Latency) were not recorded on custom test sets.	Conduct benchmark evaluations across standard defense datasets (e.g. thermal/night vision border feeds).
Local Prototype Blockchain	The ledger is stored as a single-node JSON file (`blockchain_ledger.json`) without P2P nodes or consensus.	Upgrade to a permissioned Hyperledger Fabric network with multi-node consensus.
Camera Angle & Lighting Sensitivity	Prototype trained primarily on standard RGB daylight video feeds.	Fine-tune YOLO11 models on Thermal/Infrared (FLIR) datasets for night surveillance.
Static Polygon Definition	Restricted zone polygon is configured statically in code/notebook.	Build an interactive canvas UI component allowing security operators to draw custom polygon zones directly on live video feeds.